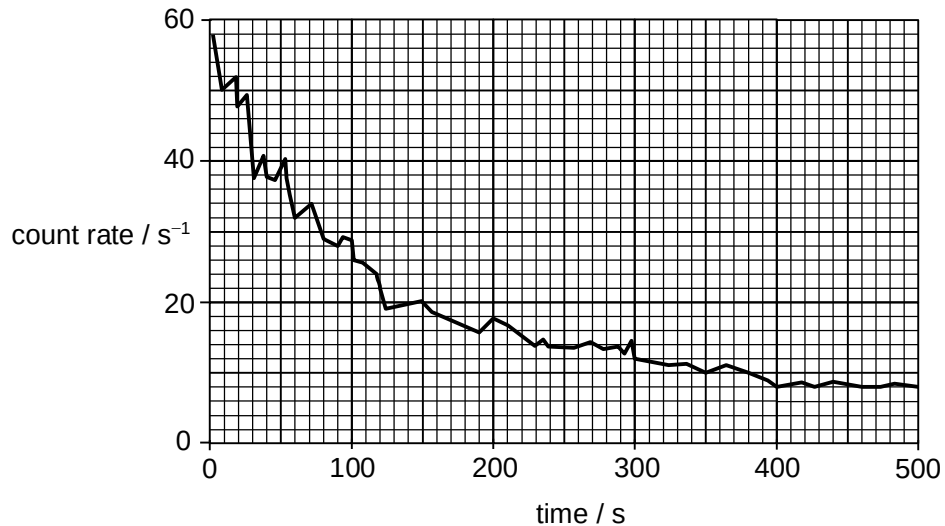


(Sufficient to only consider nucleon number in this case.)

- 30 An experiment is carried out in which the count rate is measured at a fixed distance from a sample of a certain radioactive material. The figure below shows the variation of count rate with time.



What is the approximate half-life of the material?

A 60 s

B 80 s

C 100 s

D 120 s

Ans: A

Considering background count rate to be 8 counts s⁻¹,

Half life is when count rate decreases from 58 (=50+8) to 33 (=25+8).

Best option is 60 s.